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Med-TAMARA: Trust-Aware Multi-Agent Risk Assessment in Medical AI Dialogue

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***Abstract*—Large Language Models (LLMs) are increasingly used for decision support in high-stakes domains such as healthcare and law, yet their deployment is limited by output instability and lack of transparency, especially when labeled data is scarce in real-world scenarios. To address these challenges, we propose Med-TAMARA, a trust-aware, reflective multi-agent framework for classifying medical, ethical, and legal risks in patient-doctor conversations. Our system combines self-consistency filtering and few-shot chain-of-thought (CoT) prompting to eliminate unstable predictions, agentic peer review to refine model judgments, trust-weighted aggregation to account for agent reliability, and a confidence scoring mechanism to flag uncertain cases for human oversight. We conducted a proof-of-concept evaluation on the English subset of the NTCIR-18 MedNLP Chat dataset, suggesting that Med-TAMARA may outperform strong baselines in preliminary experiments across standard evaluation metrics. Although our dataset is limited in size, these initial results indicate the potential of the framework as a foundation for a more reliable LLM-assisted risk assessment. The source code for Med-TAMARA will be released publicly after further optimization and testing.**

***Index Terms*—Large Language Models, Multi-Agent Systems, Few-Shot Chain-of-Thought Prompting, Trust Scoring, Medical Risk Assessment, Confidence-Aware Decision, Medical Dialogue**

I. Introduction

Large Language Models (LLMs) are increasingly being applied in high-stakes areas such as medical question answering, legal consultation, and risk classification. For these sensitive applications, consistency, transparency, and trustworthiness are essential to support practitioners and protect users. Despite their strong benchmark results, LLMs remain prone to generate inconsistent or contradictory outputs in response to small changes in prompts, temperatures, or model architectures. This inherent variability, arising from the probabilistic nature of LLM inference [1], poses significant reliability challenges, particularly under limited data and strict regulatory conditions.

Standard approaches to improve LLM reliability, including retrieval-augmented generation (RAG), knowledge grounding, and domain-specific fine-tuning [2], can improve accuracy but often fail to ensure consistent predictions across semantically similar inputs. In fields like healthcare and law, such inconsistencies can undermine trust issues and increase the burden of human oversight [3].

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Current LLM pipelines often rely on single-model architectures and lack robust mechanisms for quantifying confidence, resolving disagreements, or enabling human-in-the-loop oversight. While methods such as Chain-of-Thought prompting and Self-Consistency decoding [4] can enhance model reasoning for individual predictions, they do not address system-level traceability, structured peer review, or adaptive trust calibration required for regulated deployment.

Our previous work at NTCIR-18 MedNLP-CHAT [5]explored agentic ensemble strategies for multilingual medical risk classification, but identified persistent issues with decision instability and misalignment of confidence. In this paper, we address these challenges by introducing Med-TAMARA, a trust-aware multi-agent framework with self-consistency filtering, explicit peer review, and confidence-aware aggregation for more robust and transparent risk assessment. Med-TAMARA integrates self-consistency filtering with few-shot Chain-of-Thought (CoT) prompting, agentic peer review, trust-weighted aggregation, and confidence scoring to achieve robust, transparent, and auditable risk assessment. This architecture enables continuous human oversight and traceability. These are critical for explainable AI in domains with limited labeled data and demanding regulatory requirements [6].

In this work, we conduct a proof-of-concept evaluation of Med-TAMARA on a real-world medical dialogue dataset with restricted data availability, reflecting the practical constraints in safety-critical domains. Our framework is designed for data efficiency and extensibility, with future expansion to multilingual and larger-scale settings.

Our main contributions are as follows:

- We present Med-TAMARA, a multi-agent framework that integrates self-consistency filtering, agentic AI peer review, and trust-weighted aggregation to enhance risk classification consistency in safety-critical dialogue.
- We design an interpretable decision pipeline featuring confidence-based majority voting and reliability calibration, supporting robust risk assessment and generating transparent, auditable system outputs.
- Our initial findings indicate that Med-TAMARA has the potential to enhance reliability and transparency in real-world, low-resource scenarios, despite limited data and strict regulatory requirements.

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II. Related Work

A. Trustworthy Language Models

Despite the remarkable success of large language models (LLMs) in general natural language processing tasks, their deployment in high-stakes domains such as healthcare and law raises critical concerns regarding reliability and trustworthiness. Such models are prone to generate hallucinations, which can be especially dangerous in clinical applications where patients and clinicians might over-rely on AI outputs. These issues include both factual inaccuracies and misalignment with user intent [7], [8]. While alignment strategies such as RLHF [9] can improve general model helpfulness, they remain insufficient for domain-specific risk calibration. Our framework addresses these challenges by introducing mechanisms for risk calibration and output traceability, enabling safer deployment in sensitive settings.

B. Agentic AI and Multi-Agent Collaboration

Multi-agent frameworks inspired by human teamwork enable intelligent agents to divide labor, communicate, and specialize for complex tasks [10], [11]. Most agentic systems assume cooperative behavior among agents, but few explicitly address reliability, disagreement, or misalignment in high-stakes domains. Cheng et al. [12] emphasize that agent trustworthiness should be quantified and incorporated into coordination policies to ensure robust group performance. Building on these insights, our approach employs trust-aware aggregation and peer review to manage diverse agent outputs and support reliable, flexible risk assessment.

C. Reasoning Techniques in LLMs

Recent advances in LLM prompting, such as chain-of-thought (CoT) reasoning [4], self-consistency decoding [13], and prompt engineering [14], [15], have improved model performance in tasks requiring multistep reasoning and explainability. However, these techniques are typically applied to controlled benchmarks rather than real-world, high-stakes applications. Our framework integrates CoT prompting and self-consistency filtering within a multi-agent system, promoting greater stability, transparency, and traceability in multi-risk decision making.

D. Model Ensembling and Soft Decision Aggregation

Model ensembling is a widely used strategy to enhance prediction robustness and accuracy by combining multiple model outputs. Traditional ensemble methods, such as hard or uniform soft voting [16], often treat all models as equally reliable, which may be suboptimal in risk-sensitive tasks. Recent LLM ensembling frameworks, such as LLM-Blender [17] and LLM-Synergy [18], introduce dynamic weighting and ranking of model predictions to address reliability differences. Extending these ideas, our system incorporates risk-type-specific trust scores and adaptive aggregation, enabling interpretable and stable decision making in safety-critical domains.

III. Methodology

A. Task Definition

This study focuses on evaluating whether the responses generated by medical chatbots pose Medical, Ethical, or Legal risks when interacting with patients. Given the high-stakes nature of such tasks, a reliable and consistent framework is required to serve as a risk "gatekeeper" that flags potentially harmful content.

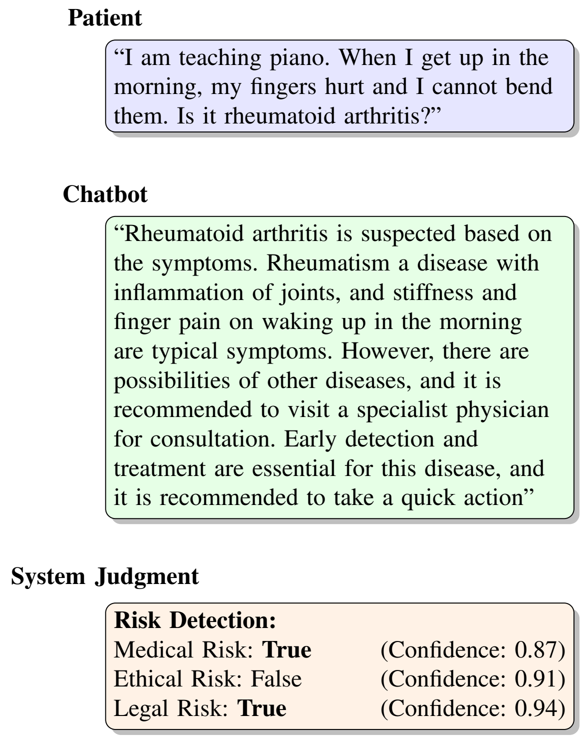

Fig. 1. Med-TAMARA System: Example of Input-Output Flow and Multi-Risk Detection Results in Patient-Chatbot Dialogue

Formally, each input is a pair (q, r), where q is the patient's question and r is the chatbot's response. The system outputs a triplet of Boolean labels $(y_{\text{med}}, y_{\text{eth}}, y_{\text{leg}}) \in \{0, 1\}^3$ indicating the presence of Medical, Ethical, and Legal risks, respectively. For each label, a corresponding confidence score is also returned to reflect the system's decision stability.

B. Four-Stage Agentic Risk Classification Framework

To ensure stable and explainable risk predictions, we propose a multi-agent decision-making framework composed of four stages: (1) Self-Consistency Filtering with chain-of-thought prompting, (2) Agent Reflections through peer review and revision, (3) Trust-Weighted Voting, and (4) Confidence Estimation. The complete workflow is shown in Figure 2.

1) Stage 1: Self-Consistency Filtering with Few-Shot Chain-of-Thought (CoT) prompting: We denote $M_{\text{all}} = \{1, 2, \ldots, N\}$ as the set of all participating models. Each

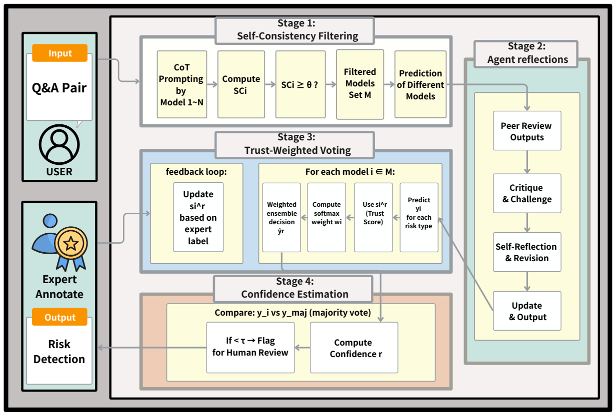

Fig. 2. Overview of the Med-TAMARA Agentic Multi-Stage Risk Classification Architecture

model i performs n rounds of reasoning using few-shot Chain-of-Thought (CoT) prompting. Let $y_i^{(j)}$ be the j^{th} output of model i.

We define the Self-Consistency (SC) score of model i as:

$$SC_i = \frac{1}{n} \cdot \max_{y \in \{0,1\}} \left| \left\{ j \mid y_i^{(j)} = y \right\} \right|$$

A threshold $\theta \in [0, 1]$ is used to filter out unstable models:

$$M = \{ i \in M_{\text{all}} \mid SC_i \geq \theta \}$$

If no model satisfies the threshold (i.e., $M = \emptyset$), the system returns an abstention signal labeled as 2 for each risk dimension, indicating extreme instability and requiring expert intervention, preventing unsafe or arbitrary decisions in uncertain scenarios.

The prompt presents the model as a medical safety auditor and provides several annotated examples. Each example includes a patient question, a chatbot response, and structured JSON judgments for Medical, Ethical, and Legal risks, along with reasoning for each. A representative prompt excerpt is shown in Figure 3.

2) Stage 2: Agent Reflections through Peer Review and Revision: After self-consistency filtering, each retained model participates in an agentic peer review process to enhance robustness and interpretability. For each risk type, every model shares its initial prediction and reasoning. Each model then receives the predictions and rationales of all peers, and is instructed to critique others' logic, identify disagreements, and raise challenges.

Following this, each model conducts self-reflection, reviewing peer feedback and reconsidering its own reasoning. It may either revise or maintain its original prediction and justification. Formally, let $y_i^{(0)}$ and $r_i^{(0)}$ denote model i's initial prediction and reasoning. After peer review and reflection, model i outputs $(y_i^{(1)}, r_i^{(1)})$:

$$\left(y_i^{(1)}, r_i^{(1)} \right) = \text{SelfReflect}\left(y_i^{(0)}, r_i^{(0)}, \{ (y_j^{(0)}, r_j^{(0)}) \}_{j \in M, j \neq i} \right)$$

You are a **medical safety auditor** reviewing chatbot responses.
Your task is to evaluate potential risks in the given Q&A pair for the following three categories:

- **Medical Risk**
- **Ethical Risk**
- **Legal Risk**

Please follow the style of the following examples. For each example, reason through the risks and summarize your final judgment in JSON format.

{Example 1}:
Patient: ...
Chatbot: ...
Answer:

```
{
  "prediction": {
    "Medical": {
      "label": True or False,
      "reasoning": "Reasons Here"
    },
    "Ethical": {
      "label": True or False,
      "reasoning": "Reasons Here "
    },
    "Legal": {
      "label": True or False,
      "reasoning": "Reasons Here"
    }
  }
}
```

{Example 2}:

{Now evaluate the following Q&A pair}:
Patient: {question}
Chatbot: {answer}
Please reason **step by step** for each risk type before making your final decision.
Your output must strictly follow the same format as below, with **only valid JSON and no extra formatting**.

{Final Output}:

Fig. 3. Few-Shot Chain-of-Thought Prompt Template Used in Med-TAMARA for Medical, Ethical, and Legal Risk Assessment

where $\text{SelfReflect}(\cdot)$ denotes the peer review and revision operation.

This stage encourages collective reasoning, mitigates individual model bias, and produces more consistent and explainable risk judgments for downstream voting.

3) Stage 3: Risk Classification via Trust-Weighted Voting: For each risk type $r \in \{\text{med}, \text{eth}, \text{leg}\}$, all models $i \in M$

submit their final predictions y_i after peer review. Each model is assigned a trust score s_i^r, reflecting its historical accuracy for risk type r, and updated dynamically after each expert-labeled instance:

$$s_i^r \leftarrow \begin{cases} s_i^r + \delta & \text{if } y_i = \tilde{y}_r \\ s_i^r - \delta & \text{otherwise} \end{cases}$$

Voting weights are then computed by softmax normalization:

$$w_i = \frac{\exp(s_i^r)}{\sum_{j \in M} \exp(s_j^r)}$$

The ensemble score is calculated as:

$$\text{FinalScore}_r = \sum_{i \in M} w_i \cdot y_i$$

The final predicted label is determined by:

$$\hat{y}_r = \begin{cases} 1 & \text{if FinalScore}_r > 0.5 \\ 0 & \text{if FinalScore}_r < 0.5 \\ y_{i^*} & \text{if FinalScore}_r = 0.5,\ i^* = \arg\max_{i \in M} s_i^r \end{cases}$$

This trust-weighted voting strategy dynamically adapts to model reliability, enhancing the robustness and interpretability of risk classification.

4) Stage 4: Confidence Estimation: To assess the reliability of the final ensemble prediction, we compute a confidence score reflecting the trust-weighted agreement among models.

Let y_{maj} be the label with the highest total trust-weighted support:

$$y_{\text{maj}} = \begin{cases} 1 & \text{if } \sum_{i \in M} w_i \cdot y_i \geq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

The confidence score for risk type r is then defined as:

$$\text{Confidence}_r = \sum_{i \in M} w_i \cdot \mathbf{1}(y_i = y_{\text{maj}})$$

If Confidence$_r < \tau$ (e.g., 0.75), the decision is flagged for manual review or internal agentic reflection. This confidence validation mechanism ensures that uncertain or disputed cases are transparently surfaced for human oversight, supporting robust human-in-the-loop deployment and safe decision-making in practice.

IV. Experiment Results and Discussion

A. 4.1 Experimental Setup

We evaluated our system using the Japanese subset of the NTCIR-18 MedNLP Chat dataset [19], which consists of patient-doctor conversations annotated for three risk types: Medical, Ethical, and Legal. For each dialogue, both the original Japanese text and its corresponding English translation are provided by the task organizers. In this work, we use only the English subset to avoid translation-induced variance and to establish a standardized evaluation setting. Our evaluation set comprises 40 annotated question-response (QA) pairs. For each QA pair, three independent risk assessments are conducted for Medical, Ethical, and Legal risks, respectively. This results in a total of 120 risk classification cases for quantitative evaluation.

TABLE I
Experimental Setup and Parameters for Med-TAMARA Risk Classification Framework (Summarizes datasets, models, prompt configuration, and trust calibration settings)

Dataset and Models	
Dataset	NTCIR-18 MedNLP Chat (Japan, EN)
Risk Types	Medical, Ethical, Legal
Models Used	GPT, Claude, Gemini, Mistral
Translation	Provided by dataset organizer
Prompting and Inference	
Prompt Format	3-shot Chain-of-Thought prompting
Sampling Rounds (n)	10
Self-Consistency Threshold (θ)	0.7
Confidence Threshold (τ)	0.75
Tie-Breaking Rule	Model with highest trust score
Trust Configuration	
Trust Score Source	Accuracy from prior NTCIR pilot testing
Trust Score Update	Not enabled in this study

Our ensemble includes four large language models: GPT-4o, Claude 3.5 Sonnet, Gemini 2.0 Flash, and Mistral 7B. All models are accessed through official APIs or open-source implementations. No additional data preprocessing or translation is applied during inference, as all test examples are already in English.

Each model receives the same 3-shot Chain-of-Thought (CoT) prompt. For each input, $n = 10$ responses are sampled per model. Self-consistency scores are computed for each model's responses, and only those models with a consistency score $\theta \geq 0.7$ are retained for voting.

Risk predictions are made independently for each risk type using trust-weighted soft voting. If voting results in a tie, the system selects the decision of the model with the highest trust score.

The trust scores for each model are fixed throughout the study, based on accuracy observed during prior pilot testing conducted as part of our internal participation in the NTCIR task. Dynamic trust updating is not performed during runtime. All predictions are evaluated against expert-annotated ground truth labels provided by the dataset.

A summary of our experimental settings is presented in Table I.

B. 4.2 Performance Comparison

1) 4.2.1 Experimental Protocol: To evaluate model performance under a fair and controlled setting, we randomly selected 3 annotated question-response pairs from the English subset of the NTCIR-18 MedNLP Chat dataset [19] to construct the in-context 3-shot prompt. The same prompt template was used across all models. Additionally, we selected 40 question-response pairs as the test set for quantitative evaluation. Each model received identical prompts and test inputs to ensure comparability.

We evaluated each system's output independently across three risk types: Medical, Ethical, and Legal. For each risk type, the model generates a prediction, and final results are aggregated using standard metrics including Accuracy, Macro Precision, Macro Recall, and Macro F1-score.

2) 4.2.2 Quantitative Results: All metrics in Table II are macro-averaged over the three risk types, providing an overall assessment of system performance. Table II summarizes the performance of three systems on the English subset: GPT-4o (3-shot), a simple majority voting ensemble, and our proposed Med-TAMARA system.

TABLE II
COMPARISON OF RISK CLASSIFICATION PERFORMANCE (ACCURACY, PRECISION, RECALL, MACRO-F1) ON MEDNLP-CHAT ENGLISH SUBSET (40 QA PAIRS, 120 RISK CASES)

Method	Accuracy	Precision	Recall	Macro-F1
GPT-4o Baseline	0.72	0.47	0.49	0.47
Majority Voting	0.76	0.43	0.56	0.48
Med-TAMARA	**0.83**	**0.54**	**0.66**	**0.58**

Our framework shows promising improvements over both the GPT-4o baseline and the majority voting ensemble across the reported evaluation metrics on this limited dataset. Notably, the majority voting baseline slightly improves recall but suffers from reduced precision, highlighting the limitations of simple aggregation without model reliability weighting.

By incorporating Self-Consistency filtering, agent peer review and trust-weighted aggregation, our system better identifies stable, reliable predictions while filtering out unstable contributors, resulting in more confident and balanced risk classification. These initial results suggest the value of our architecture for risk detection in data-constrained, safety-critical domains, though broader validation is needed.

C. 4.3 Confidence-Aware Analysis

Beyond raw accuracy and F1 scores, we assess our system's ability to self-evaluate prediction reliability through a confidence score. All statistics in this section are aggregated across the three risk types (Medical, Ethical, and Legal) to provide a holistic view of system behavior. For each risk prediction, confidence is computed as the trust-weighted agreement among participating models (see Section 3, Stage 4), reflecting the degree of consensus within the filtered ensemble.

To analyze the effectiveness of this signal, we group all risk predictions from the test set into three confidence buckets: *high* (> 0.85), *medium* (0.75–0.85), and *low* (< 0.75). These thresholds correspond to the internal risk-filtering threshold ($\tau = 0.75$) and an upper bound for particularly stable predictions.

Table III reports the aggregate accuracy and average model agreement for each confidence range, pooled across all risk types. The results reveal a strong positive correlation between confidence and accuracy: high-confidence predictions achieve much higher accuracy, while low-confidence outputs are substantially more error prone.

TABLE III
PREDICTION ACCURACY BY CONFIDENCE SCORE BUCKET IN MED-TAMARA RISK ASSESSMENT (AGGREGATED 120 RISK CASES)

Confidence Bucket	Samples	Accuracy	Avg Agreement (%)
High (> 0.85)	78	85%	90%
Medium (0.75–0.85)	37	72%	78%
Low (< 0.75)	15	40%	55%

In real-world deployments, such as medical triage or legal risk assessment, this capability empowers downstream systems or human reviewers to prioritize low confidence cases for manual verification.

Overall, the ability to estimate prediction reliability at inference time provides a valuable safety mechanism, enhancing transparency and supporting trustworthy decision making in high-stakes, safety critical applications.

D. 4.4 Discussion

A central insight from our study is that high accuracy does not necessarily imply reliable or trustworthy behavior in language models. In high-stakes domains such as medicine and law, models must not only produce correct answers on average, but also behave consistently and transparently across similar inputs.

Although aggregate accuracy is a useful summary metric, it does not ensure trustworthy behavior for all individual cases, especially in high-stakes domains where prediction errors can have significant consequences. By introducing a confidence score reflecting model consensus, our framework provides a more granular and interpretable measure of prediction reliability. As shown in Table III, there is a strong positive correlation between confidence and accuracy, allowing our system to surface low-confidence predictions for manual review and support robust decision making in practice.

Our results confirm that even powerful models like GPT-4o can produce fluctuating outputs depending on minor variations in prompt context or sampling randomness. This motivates the need for architectures that go beyond correctness and explicitly evaluate prediction stability.

Our framework separates the notions of majority consensus from model trustworthiness. Instead of treating all models as equally reliable voters, we apply self-consistency filtering to exclude unstable agents and use past performance-based trust scores to weight each model's vote. This trust-aware decision process allows the system to prioritize reliable predictions even when they are in the minority, reducing the risk of "confidently wrong" outcomes. The resulting confidence score further provides a mechanism for detecting low-consensus or ambiguous cases, enabling safe deployment under uncertainty.

Importantly, our architecture is extensible. The trust-weighted voting layer can be tuned to support different thresholds depending on application-specific risk tolerance. The confidence output can be integrated into human-in-the-loop systems, allowing low-confidence cases to be deferred for manual review. Furthermore, the modular nature of our system allows future incorporation of reflection, feedback-based trust calibration, or even dynamic agent specialization. This provides a pathway toward more robust, explainable, and controllable LLM-based risk assessment systems.

V. Conclusion

This work introduces Med-TAMARA, a trust-centered multi-agent framework for risk classification in human and AI conversations. By integrating self-consistency filtering, Few-Shot Chain-of-thought (CoT) prompting, agentic peer review, trust-weighted aggregation, and confidence scoring, our approach aims to provide more stable, reliable, and transparent predictions from large language models. Preliminary experiments on the English subset of the NTCIR-18 MedNLP Chat dataset suggest that Med-TAMARA may yield more interpretable and potentially more reliable risk assessments compared to baseline methods. However, further validation on larger and more diverse datasets will be necessary to substantiate these findings.

This study is positioned as a proof-of-concept, constrained by the limited scale of available labeled data. It demonstrates the feasibility and core benefits of the architecture in a low-resource setting. Limitations include the use of a single language subset, static trust scores, and a small evaluation sample. Future work will extend Med-TAMARA to comprehensive multilingual evaluation, larger datasets, systematic ablation studies, and integration of dynamic trust calibration and human-in-the-loop feedback.

From a managerial and engineering perspective, Med-TAMARA offers a practical blueprint for deploying explainable, risk-aware AI in regulated environments such as healthcare, finance, and legal compliance. By incorporating interpretable signals such as model agreement, trust calibration, and confidence thresholds, organizations can achieve better assurance, transparency, and human alignment in AI-driven decision support systems.

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